in quantum mechanics there are only certain discrete results we can obtain when we measure a particle's spin.

For an electron, for example, there are two possible measurement outcomes for spin. First pick an axis with respect to which we measure the spin. We always find that the electron is spinning either clockwise or counterclockwise when we look along that axis, and always at the same rate. These are conventionally referred to as "spin-up" and "spin-down." Think of the "right-hand rule": if you wrap the fingers of your right hand in the direction of rotation, your thumb will be pointing along the appropriate up/down axis.




A spinning electron is a tiny magnet, with north and south magnetic poles, much like the Earth; the spin axis points toward the north pole. One way of measuring the spin of a particular electron is to shoot it through a magnetic field, which will deflect the electron by a bit depending on how its spin is oriented. (As a technicality, the magnetic field has to be focused in the right way—spread out on one side, pinched tightly on the other—for this to work.)

If I told you that the electron had a certain total spin, you might make the following prediction for such an experiment: the electron would be deflected up if its spin axis were aligned with the external field, deflected down if its spin were aligned in the opposite direction, and deflected at some intermediate angle if its spin were somewhere in between. But that's not what we see.

This experiment was first performed in 1922, by German physicists Otto Stern (an assistant to Max Born) and Walter Gerlach, before the idea of spin had been explicitly spelled out. What they saw was remarkable. Electrons are indeed deflected by passing through the magnetic field, but they either go up, or they go down; nothing in between. If we rotate the magnetic field, the electrons are still deflected in the direction of the field they pass through, either along or against it, but no intermediate values. The measured spin, like the energy of an electron orbiting an atomic nucleus, appears to be quantized.